QM – Course Distribution

Lecture No	Syllabus	Remarks
L1	Failure of CM, Qualitative Review of Relevant Experiments	
L2	De Broglie Waves, Experiments, XRD and TEM	
L3	Uncertainty Principle, Examples, Experimental Verification	
L4	Wave Function and Schrodinger Equation	
L5	Probability interpretation and Operator Algebra	
L6	Every Day Examples of Quantum Mechanics	
L7	Noble Prize in Physics 2022: Quantum Mechanics	
L8	**1-d SE, Time dependent and Time Independent SE**	
L9	Particle in a Box: Nano Particle – Optical Absorption/Emission	
L11	Harmonic Oscillator, Infinite Square Well, Finite Square well	

References:

1. Concepts of Modern Physics by Arthur Beiser (6th Ediion)
2. Principles of Quantum Mechanics by R Shankar

Classical Mechanics is an approximation of Quantum Mechanics

- The Fundamental difference between classical (or Newtonian) mechanics and Quantum Mechanics, lies in what they describe.
- In classical mechanics, the future history of a particle is completely determined by its initial position and momentum together with the forces that act upon it. In the everyday world these quantities can all be determined well enough for the predictions of Newtonian mechanics to agree with what we find.

- Quantum mechanics also arrives at relationships between observable quantities, but the uncertainty principle suggests that the nature of an observable quantity is different in the atomic realm. Cause and effect are still related in quantum mechanics, but what they concern needs careful interpretation. In quantum mechanics the kind of certainty about the future characteristic of classical mechanics is impossible because the initial state of a particle cannot be established with sufficient accuracy.

- The quantities whose relationships quantum mechanics explores are probabilities.

The quantities whose relationships quantum mechanics explores are *probabilities.*

Instead of asserting, for example, that the radius of the electron's orbit in a ground state hydrogen atom is always exactly 0.053 nm, as the Bohr theory does, quantum mechanics states that this is the *most probable* radius.

In a suitable experiment most trials will yield a different value, either larger or smaller, but the value most likely to be found will be 0.053 nm

- Quantum mechanics might seem a poor substitute for classical mechanics.
- However, classical mechanics turns out to be just an approximate version of quantum mechanics.

- The certainties of classical mechanics are illusory, and their apparent agreement with experiment occurs because ordinary objects consist of so many individual atoms that departures from average behavior are unnoticeable.

- Instead of two sets of physical principles, one for the macroworld and one for the microworld, there is only the single set included in quantum mechanics.

Failures of Classical Mechanics:

- In our everyday experience there is nothing mysterious or ambiguous about the concepts of **particle** and **wave.**

- A stone dropped into a lake and the ripples that spread out from its point of impact apparently have in common only the ability to carry energy and momentum from one place to another.

- Classical physics, which mirrors the "physical reality" of our sense impressions, treats particles and waves as separate components of that reality. The mechanics of particles and the optics of waves are traditionally independent disciplines, each with its own chain of experiments and principles based on their results.

- The physical reality we perceive has its roots in the microscopic world of atoms and molecules, electrons and nuclei, but in this world there are neither particles nor waves in our sense of these terms.

What is Light?

Is Light consists of Photons or Electromagnetic Waves?

We have seen dual properties of Light.

1. **Through Diffraction Experiments – Light is a wave**

2. **Through Photoelectric effect – Light is a particle**

Particle Properties of Waves

An Experiment with Waves and Particles (Classical)

There exist in classical physics two distinct entities: particles and waves.

Particles:

- Particles are localized bundles of energy and momentum. They are described at any instant by the state parameters **x** and **p** (m**v**). These parameters evolve in time according to some equations of motion.
- Given the initial values $\mathbf{x(t_i)}$ and $\mathbf{p(t_i)}$ at time t_i, the trajectory **x(t)** may be deduced for all future times from the equations of motion (usually with **F** = m**a**)

Waves:

- A wave, in contrast, is a disturbance spread over space. It is described by a wave function $\boldsymbol{\psi}$ **(x,t)** which characterizes the disturbance at the point **x** at time **t**.

- In the case of sound waves, $\boldsymbol{\psi}$ is the excess air pressure above the normal, while in the case of electromagnetic waves, $\boldsymbol{\psi}$ can be any component of the electric field vector E.

The analogs of x and p for a wave are **ψ** and **dψ/dt** at each point **r**, assuming **ψ** obeys a second-order wave equation in time, such as

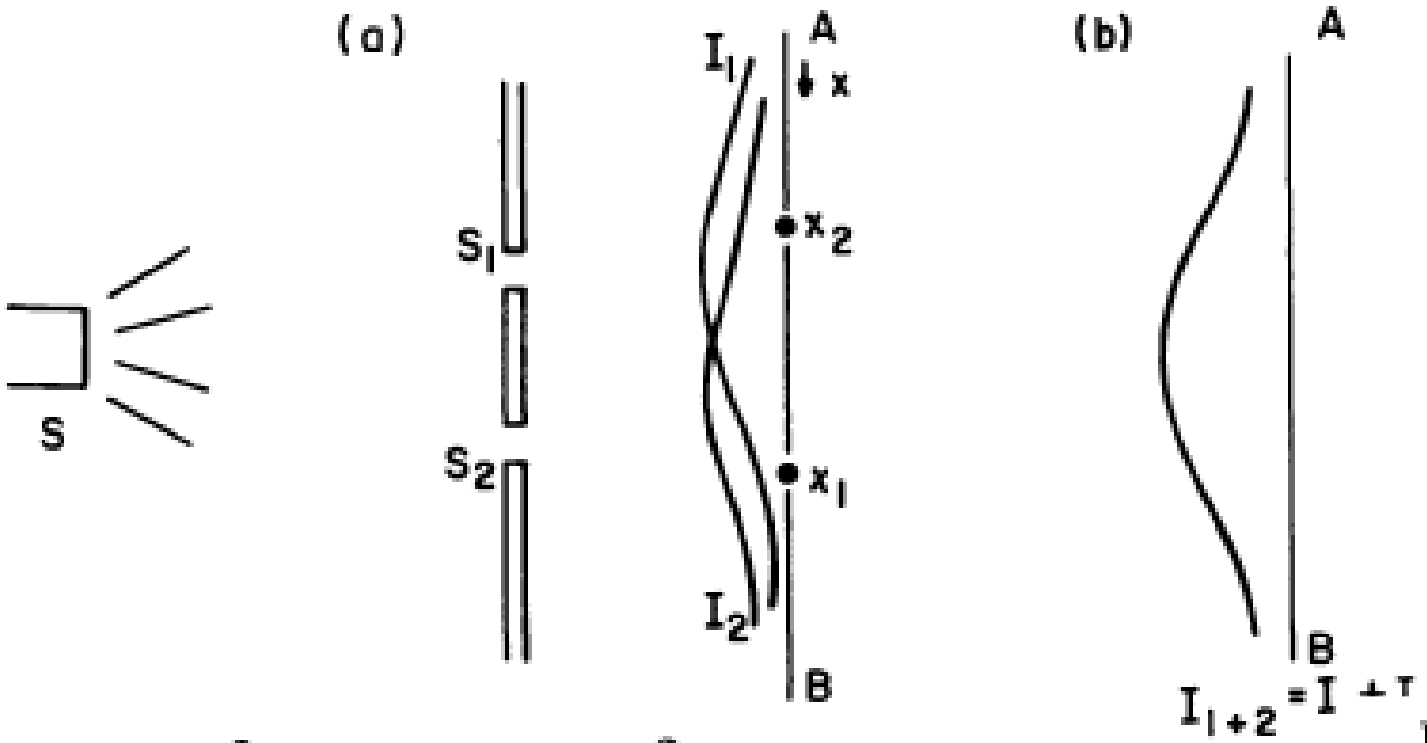
$$\nabla^2 \psi = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

which describes waves propagating at the speed of light, c. Given $\psi(\mathbf{r}, 0)$ and $\dot{\psi}(\mathbf{r}, 0)$ one can get the wave function $\psi(\mathbf{r}, t)$ for all future times by solving the wave equation.

Of special interest to us are waves that are periodic in space and time, called *plane waves*. In one dimension, the plane wave may be written as

$$\psi(x, t) = A \exp\left[i\left(\frac{2\pi}{\lambda} x - \frac{2\pi}{T} t \right) \right] \equiv A \exp[i\phi]$$

Classical Picture



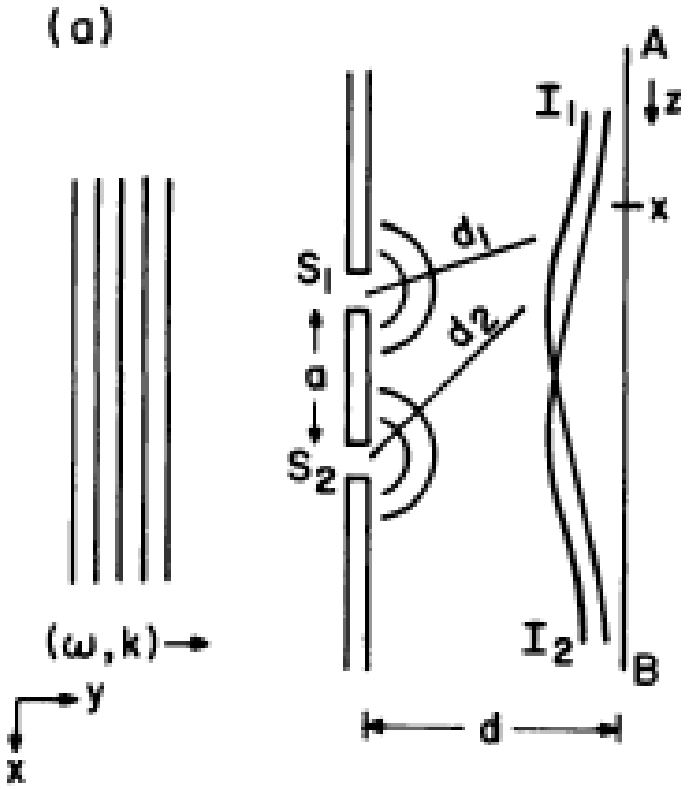
$$I_{1+2} = I_1 + I_2$$

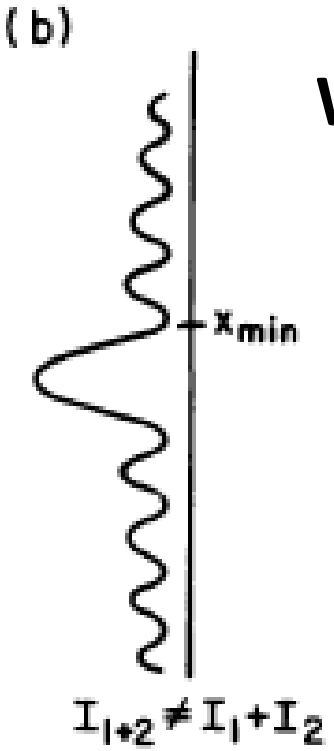
$$I_1 = |\psi_1|^2 \qquad I_2 = |\psi_2|^2$$

Figure 3.2. (a) Intensity pattern when S_1 or S_2 is open, due to a beam of incident particles. (b) The pattern with both slits open according to classical mechanics ($I_{1+2} = I_1 + I_2$).

R Shankar, Principles of QM Chapter 3

Wave Picture: Double Slit Experiment





$$I_{1+2} = |\psi_1 + \psi_2|^2$$

$$I_{1+2} \neq I_1 + I_2$$

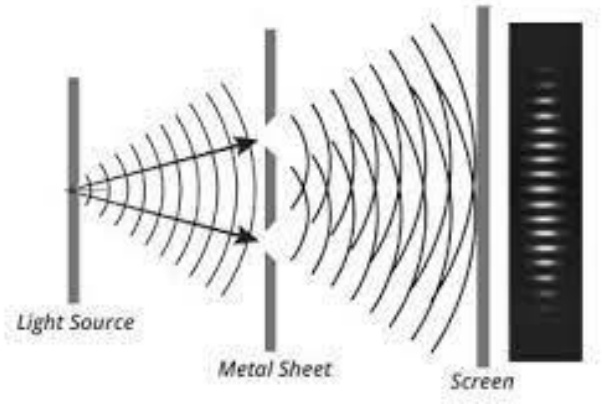


Figure 3.1. (a) When a wave $\psi = e^{i(ky - \omega t)}$ is incident on the screen with *either* slit S_1 or S_2 open, the intensity patterns I_1 and I_2, respectively, are measured by the row of detectors on AB. (b) With both slits open, the pattern I_{1+2} is observed. Note that $I_{1+2} \neq I_1 + I_2$. This is called interference.

R Shankar, Principles of QM Chapter 3

Measurement-induced decoherence and information in double-slit interference

Joshua Kincaid , Kyle McLell and Michael Zwolak

https://arxiv.org/pdf/1606.09442.pdf; American Journal of Physics 84, 522–530 (2016)

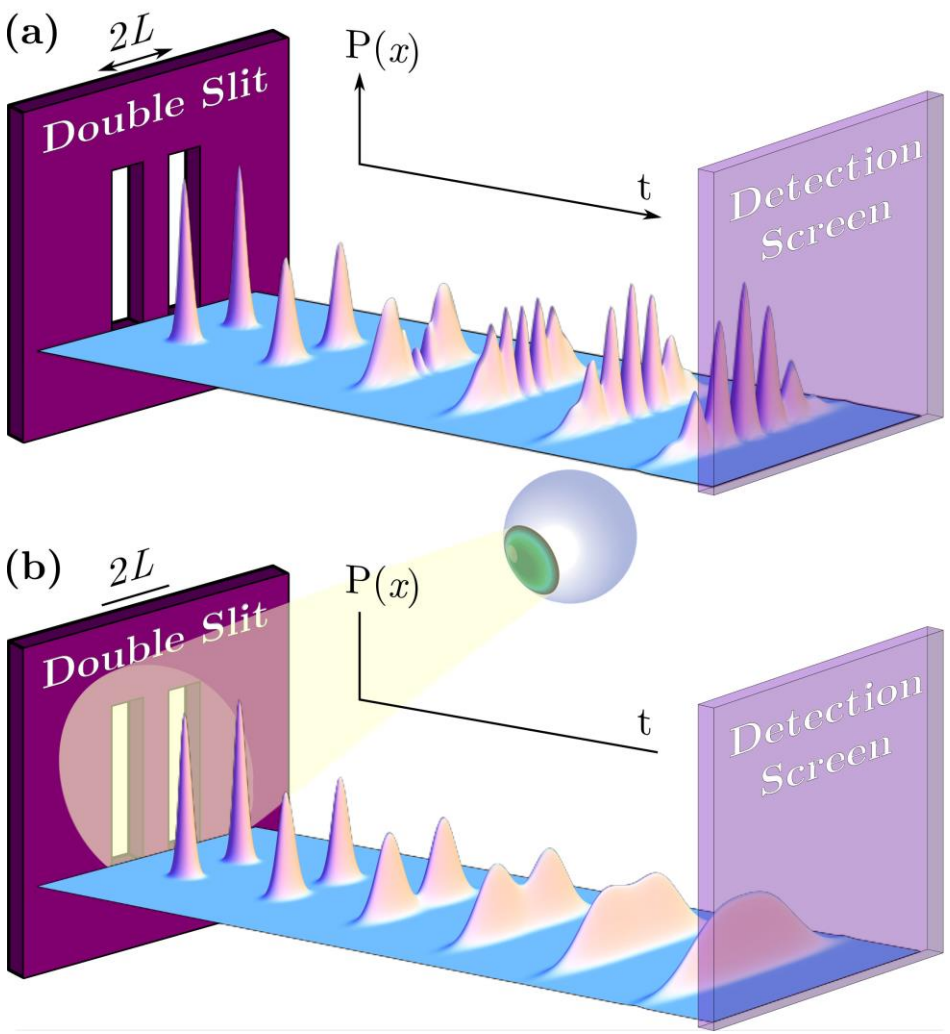


FIG. 1. Schematic representation of the one dimensional version of the double slit experiment. In this setup, a coherent superposition of two Gaussian wave packets of width emerge from the two slits separated by a distance 2L.

(a) When no measurement is made and coherence is otherwise preserved, the probability density P(x) shows that as time progresses the two packets begin to interfere, ultimately resulting in a well defined interference pattern.

(a) In the case of a perfect measurement, each particle takes either the left or the right path. In this case P(x) observed at the detection screen will be an incoherent sum of the two spreading Gaussian wave packets, i.e., no interference will be present.

The double-slit experiment is the paradigmatic example of quantum mechanical “weirdness." According to Feynman, it contains “the only mystery [of quantum mechanics]" where the particles – whatever they may be, electrons, photons, etc. - behave “sometimes like a particle and sometimes like a wave”.

In the conceptually simplest set-up, a stream of particles is incident on a barrier in which two slits have been made to allow passage. On the far side of the barrier is a detection screen, which is ultimately examined to determine the outcome of the experiment. As is well known, the detection screen will reveal an interference pattern if no attempt by an observer or otherwise, e.g., the environment was made to determine which path was taken by the particles, even when the particles are sufficiently separated in time that only one is ever in the region containing the barrier and the screen.

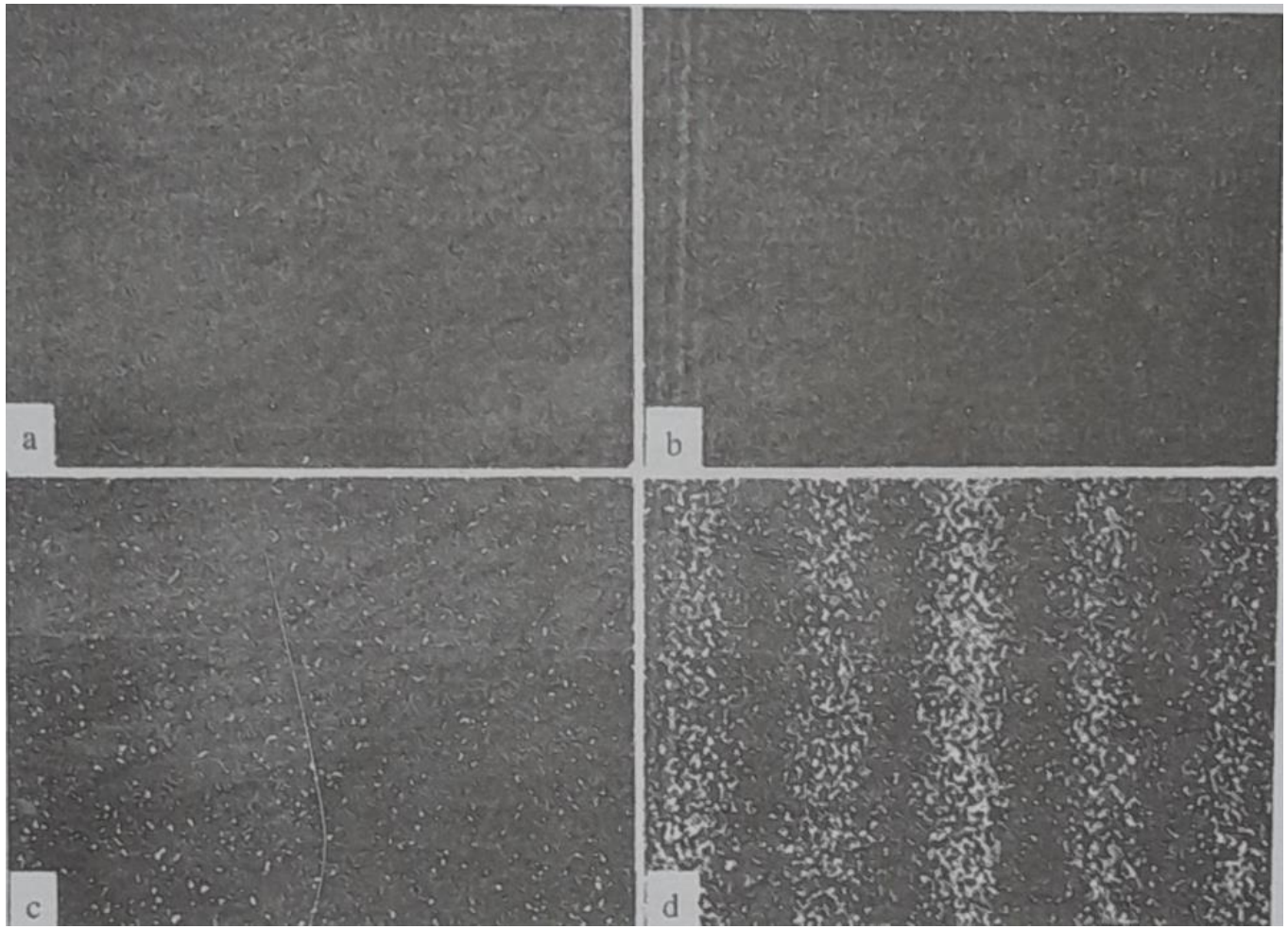
The interference pattern disappears if a detection apparatus is placed to determine which path is taken by each particle. In other words, acquiring a particle's “which-path" information prevents it from exhibiting interference. If however, one is willing to forgo perfect determination of the path, partial interference can be preserved.

ELECTRON INTEFERENCE:

Conventional Electron Diffraction would be same as double slit pattern.

As we decrease the intensity of the electron beam, until only one electron is present in the apparatus at any particular time, according to the statistical interpretation each electron will produce a spot on the screen.

QM cannot predict the precise location of that spot – all it can tell us is the probability of a given electron landing at a particular place. But, if we are patient and wait for a hundred thousand electrons – one at a time – to make the trip, the accumulating spots reveal the classic two slit interference pattern.

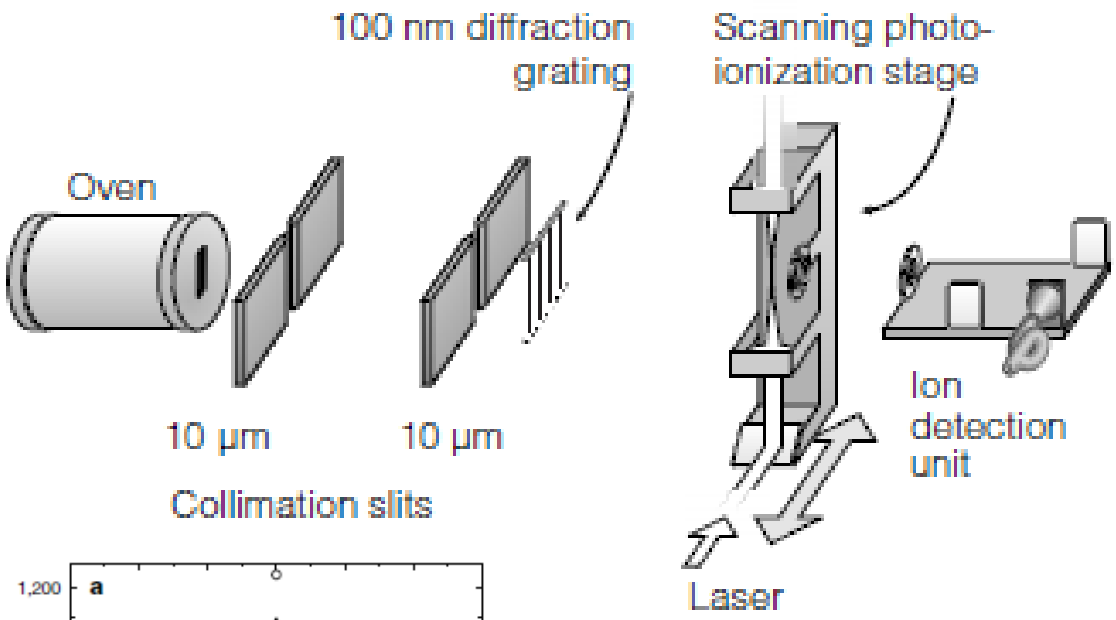


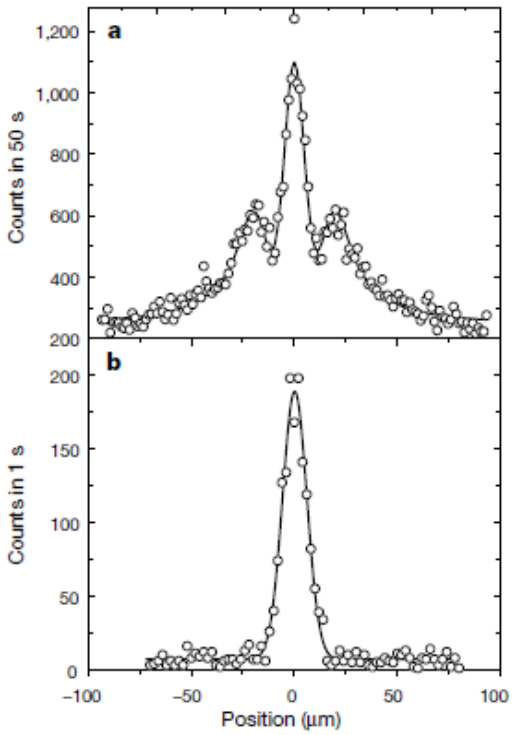
Build-up of the electron interference pattern:
(a)8 Electrons (b) 270 Electrons
(c) 2000 Electrons (d) 160,000 electrons

Wave-particle duality of C(60) molecules

M Arndt, … A Zeilinger; Nature 40 (1999) 680

Quantum superposition lies at the heart of quantum mechanics and gives rise to many of its paradoxes. Superposition of de Broglie matter waves' has been observed for massive particles such as electrons, atoms and dimers, small van der Waals clusters, and neutrons. But matter wave interferometry with larger objects has remained experimentally challenging, despite the development of powerful atom interferometric techniques for experiments in fundamental quantum mechanics, metrology and lithography. Here we report the observation of de Broglie wave interference of C(60) molecules by diffraction at a material absorption grating. This molecule is the most massive and complex object in which wave behaviour has been observed. Of particular interest is the fact that C(60) is almost a classical body, because of its many excited internal degrees of freedom and their possible couplings to the environment. Such couplings are essential for the appearance of decoherence, suggesting that interference experiments with large molecules should facilitate detailed studies of this process.





Interference pattern produced by C60 molecules.

a, Experimental recording (open circles) and fit using Kirchhoff diffraction theory (continuous line). The expected zeroth and first-order maxima can be clearly seen. Details of the theory are discussed in the text.

b, The molecular beam profile without the grating in the path of the molecules

BLACKBODY RADIATION

Only the quantum theory of light can explain its origin

The ability of a body to radiate is closely related to its ability to absorb radiation. This is to be expected, since a body at a constant temperature is in thermal equilibrium with its surroundings and must absorb energy from them at the same rate as it emits energy. It is convenient to consider as an ideal body one that absorbs *all* radiation incident upon it, regardless of frequency. Such a body is called a **blackbody.** The point of introducing the idealized blackbody in a discussion of thermal radiation is that we can now disregard the precise nature of whatever is radiating, since all blackbodies behave identically. In the laboratory a blackbody can be approximated by a hollow object with a very small hole leading to its interior.

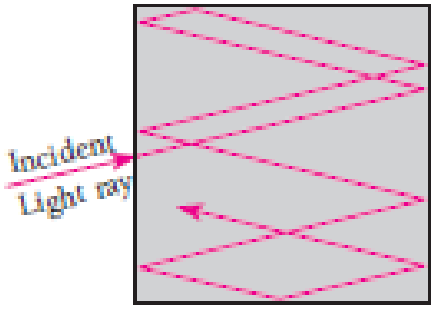
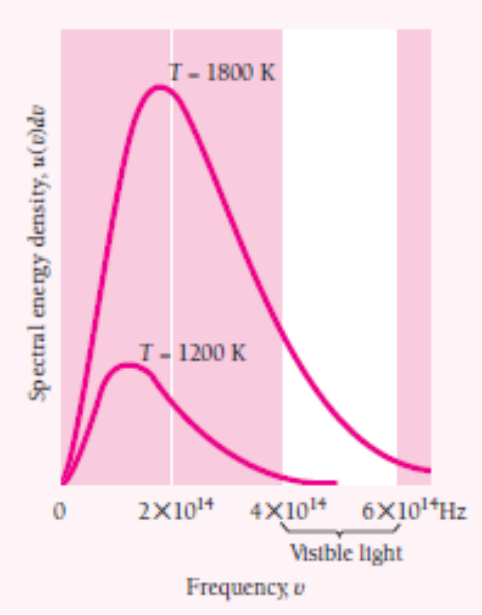
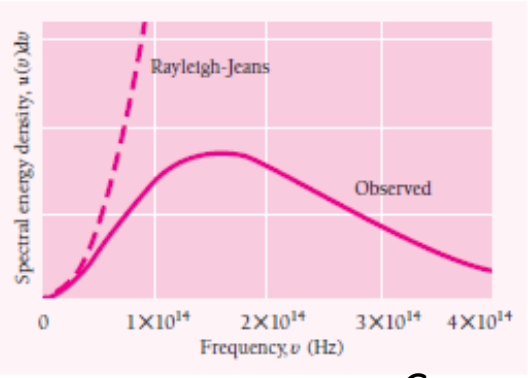


Figure 2.5 A hole in the wall of a hollow object is an excellent approximation of a blackbody.



Rayleigh-Jeans formula

$$u(\nu)\,d\nu = \bar{\epsilon}G(\nu)\,d\nu = \frac{8\pi kT}{c^3}\nu^2\,d\nu$$



Concepts in Modern Physics: Arthur Beiser (6E)

Planck Radiation Formula

In 1900 the German physicist Max Planck used "lucky guesswork" (as he later called it) to come up with a formula for the spectral energy density of blackbody radiation:

Planck radiation formula

$$u(\nu)\,d\nu = \frac{8\pi h}{c^3}\frac{\nu^3\,d\nu}{e^{h\nu/kT}-1}$$

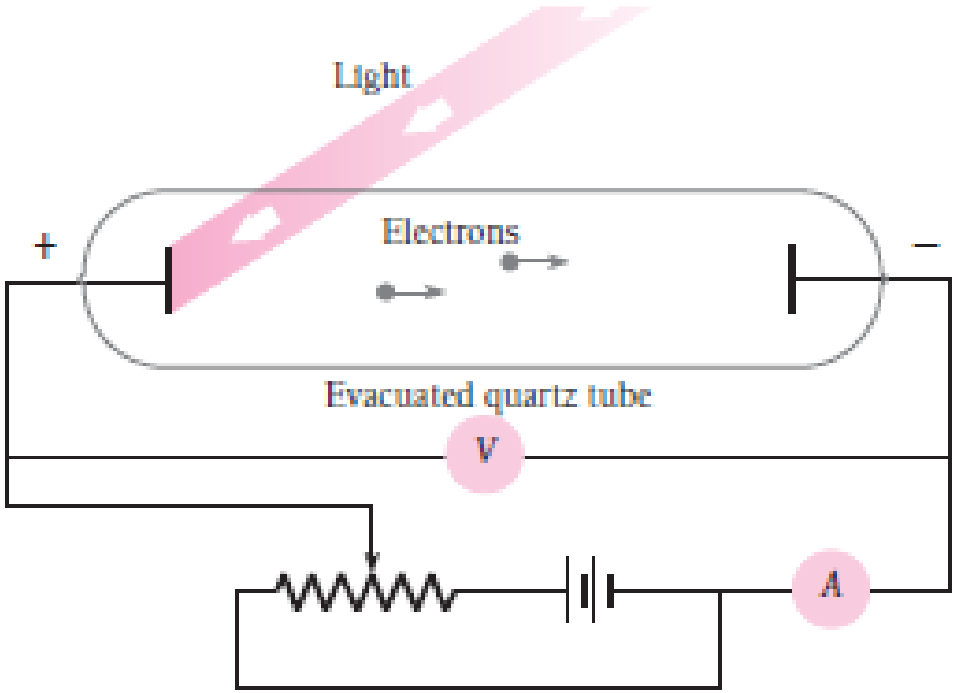
Thus at low frequencies Planck's formula becomes

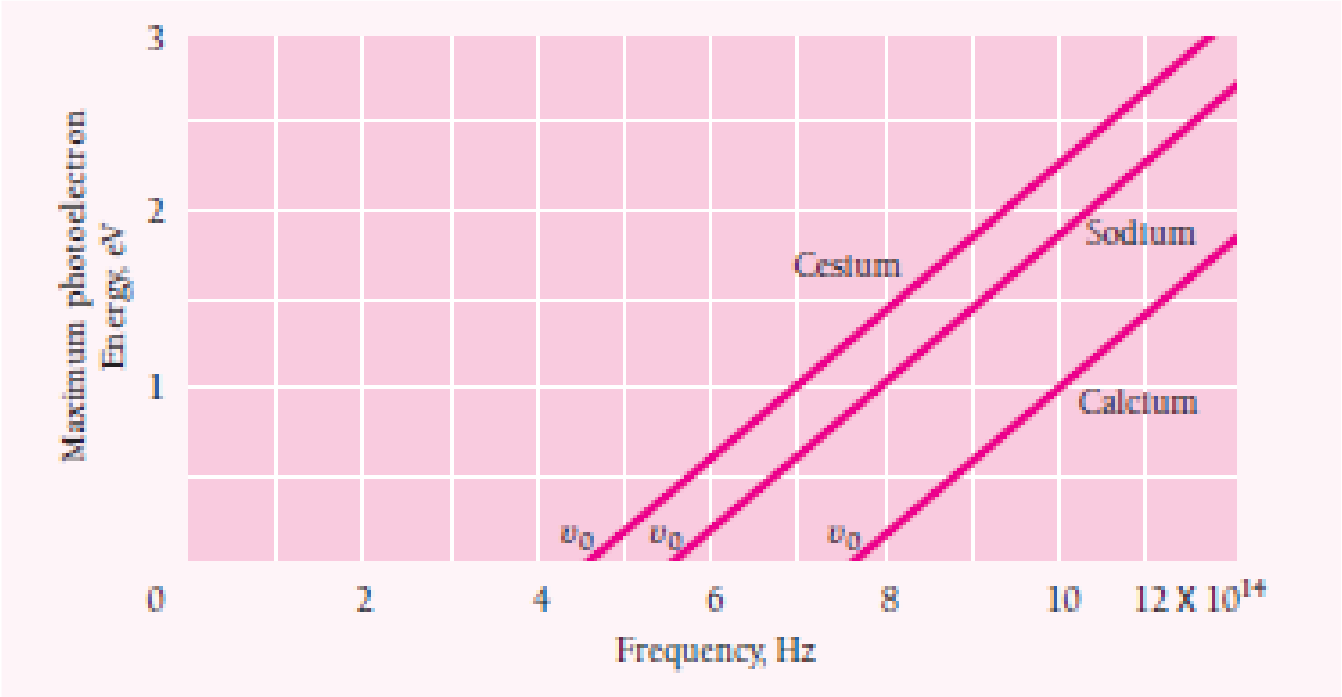
$$u(\nu)\,d\nu \approx \frac{8\pi h}{c^3}\nu^3\left(\frac{kT}{h\nu}\right)d\nu = \frac{8\pi kT}{c^3}\nu^2\,d\nu$$

Problem: Assume that a certain 660-Hz tuning fork can be considered as a harmonic oscillator whose vibrational energy is 0.04 J. Compare the energy quanta of this tuning fork with those of an atomic oscillator that emits and absorbs orange light whose frequency is 5.00×10^{14} Hz.

PHOTOELECTRIC EFFECT

The energies of electrons liberated by light depend on the frequency of the light





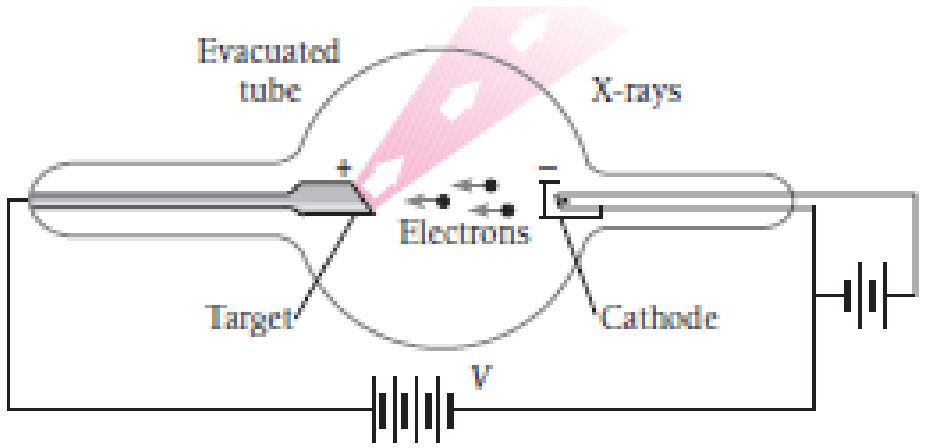
$$h\nu = KE_{max} + h\nu_0$$

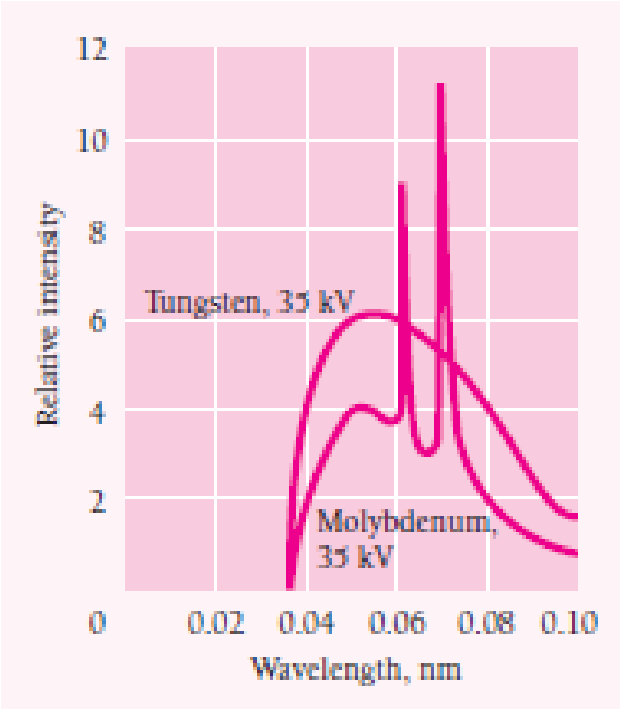
$$KE_{max} = h\nu - h\nu_0 = h(\nu - \nu_0)$$

$$h\nu = KE_{max} + \phi$$

X-RAYS

They consist of high-energy photons





$$\lambda_{min} = \frac{1.24 \times 10^{-6}}{V} \text{ V} \cdot \text{m}$$

$$Ve = h\nu_{max} = \frac{hc}{\lambda_{min}}$$

$$\lambda_{min} = \frac{hc}{Ve} = \frac{1.240 \times 10^{-6}}{V} \text{ V} \cdot \text{m}$$

X-RAY DIFFRACTION

How x-ray wavelengths can be determined

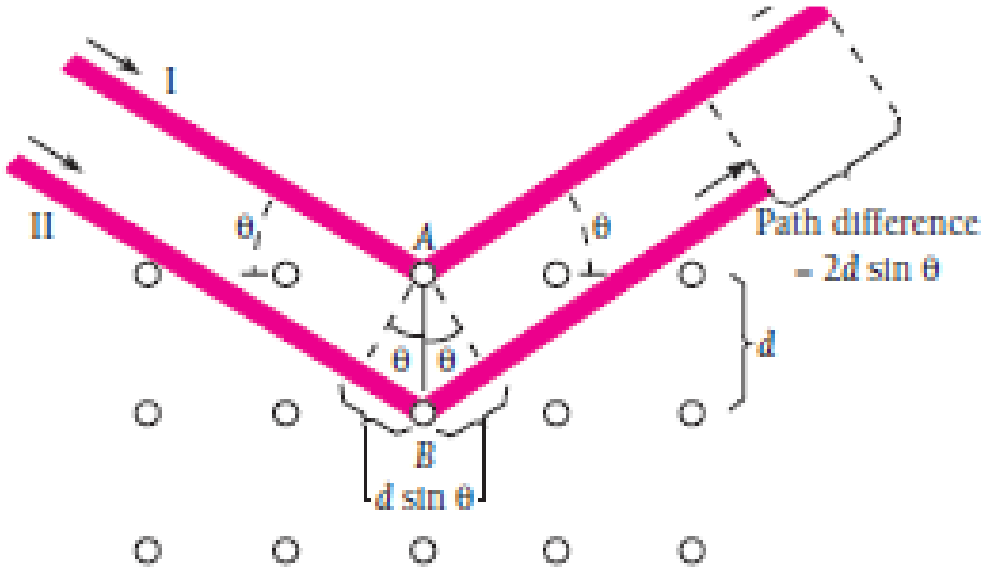


Figure 2.20 X-ray scattering from a cubic crystal.

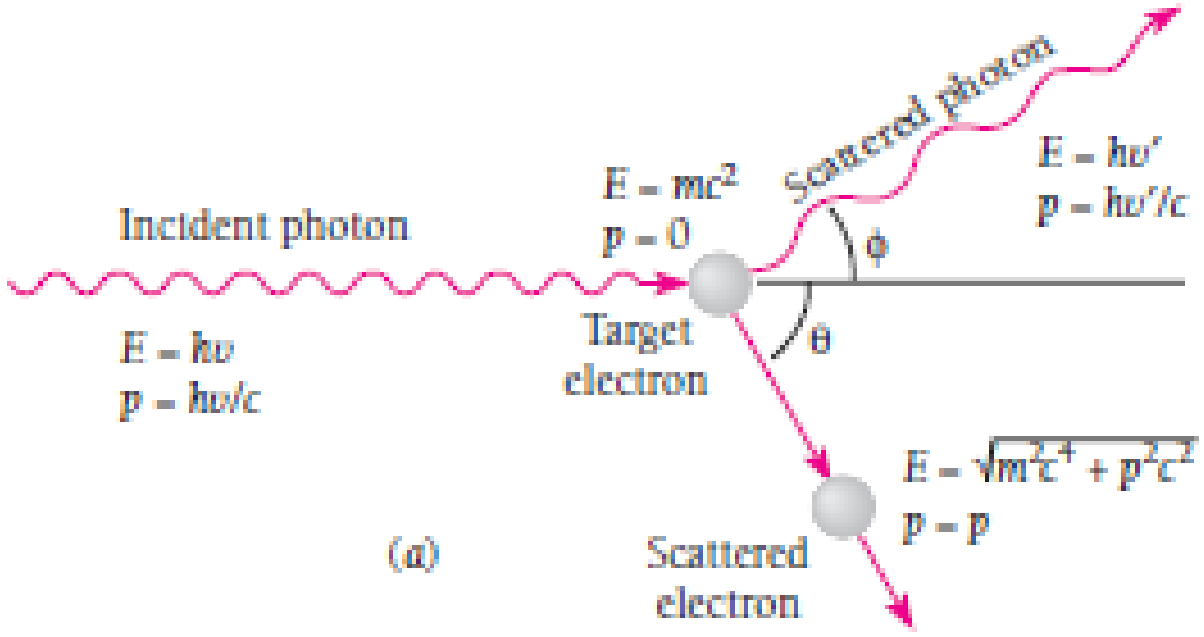
$$2d \sin \theta = n\lambda \qquad n = 1, 2, 3, \ldots$$

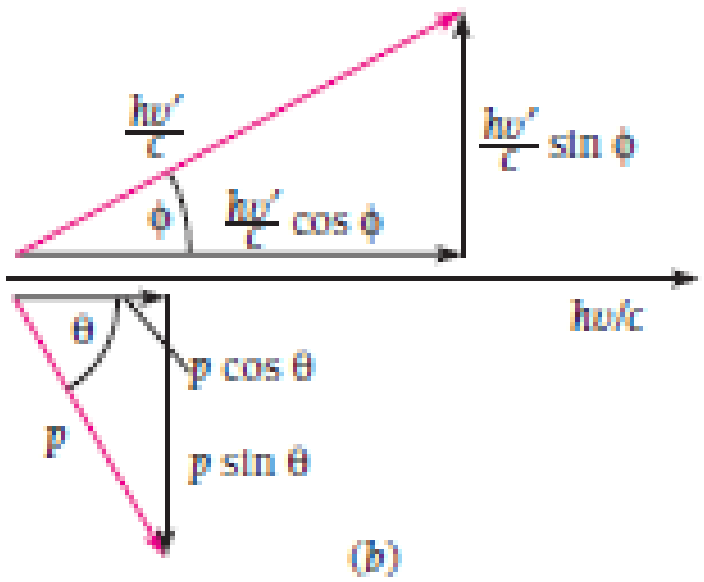
COMPTON EFFECT

Further confirmation of the photon model

Loss in photon energy = gain in electron energy

$$h\nu - h\nu' = \text{KE}$$





$$\lambda' - \lambda = \frac{h}{mc}(1 - \cos\phi)$$

3.1 DE BROGLIE WAVES

A moving body behaves in certain ways as though it has a wave nature

A photon of light of frequency ν has the momentum

$$p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

since $\lambda\nu = c$. The wavelength of a photon is therefore specified by its momentum according to the relation

Photon wavelength

$$\lambda = \frac{h}{p} \tag{3.1}$$

De Broglie suggested that Eq. (3.1) is a completely general one that applies to material particles as well as to photons. The momentum of a particle of mass m and velocity v is $p = \gamma m v$, and its **de Broglie wavelength** is accordingly

De Broglie wavelength

$$\lambda = \frac{h}{\gamma m v} \tag{3.2}$$

Find the de Broglie wavelengths of (*a*) a 46-g golf ball with a velocity of 30 m/s, and (*b*) an electron with a velocity of 10^7 m/s.

Solution

(*a*) Since $v \ll c$, we can let $\gamma = 1$. Hence

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}\ \mathrm{J \cdot s}}{(0.046\ \mathrm{kg})(30\ \mathrm{m/s})} = 4.8 \times 10^{-34}\ \mathrm{m}$$

The wavelength of the golf ball is so small compared with its dimensions that we would not expect to find any wave aspects in its behavior.

(*b*) Again $v \ll c$, so with $m = 9.1 \times 10^{-31}$ kg, we have

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}\ \mathrm{J \cdot s}}{(9.1 \times 10^{-31}\ \mathrm{kg})(10^7\ \mathrm{m/s})} = 7.3 \times 10^{-11}\ \mathrm{m}$$

The dimensions of atoms are comparable with this figure—the radius of the hydrogen atom, for instance, is 5.3×10^{-11} m. It is therefore not surprising that the wave character of moving electrons is the key to understanding atomic structure and behavior.

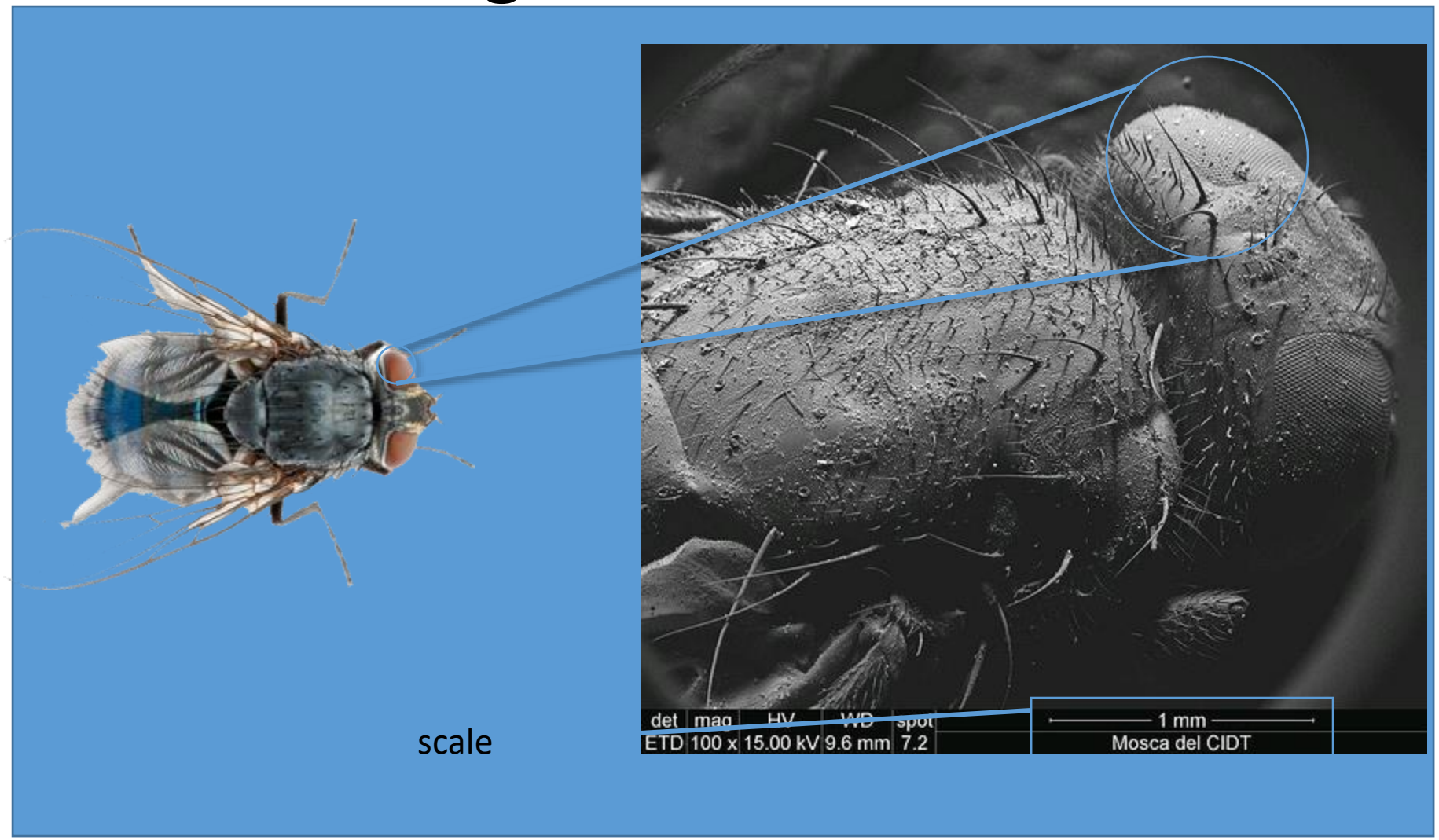
What is a Microscope?

It is an instrument for magnifying things too small to see with the naked eye

Ref: Web

Ref: Researchgate – by Beata Żuraw
Habit of the flower (a) and petal sections of Rosa gallica (SEM)

Resolution & Magnification



What is the Magnification that is required to SEE atoms?

Our Eye can see typically ≈ 0.3 mm

Size of an atom= 1 Å = 10^{-10} m

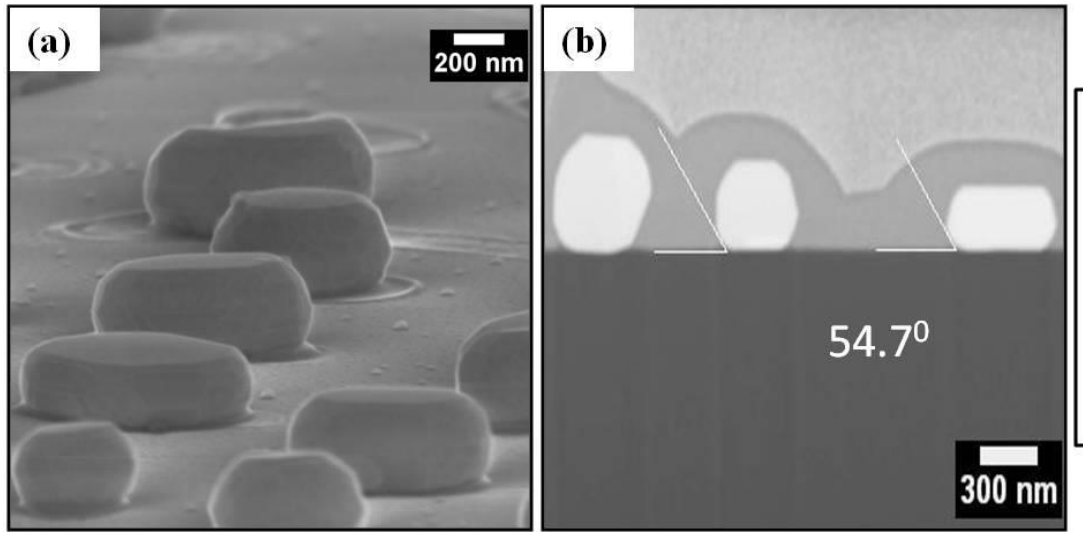
Magnification = 0.3 mm/ 1 Å

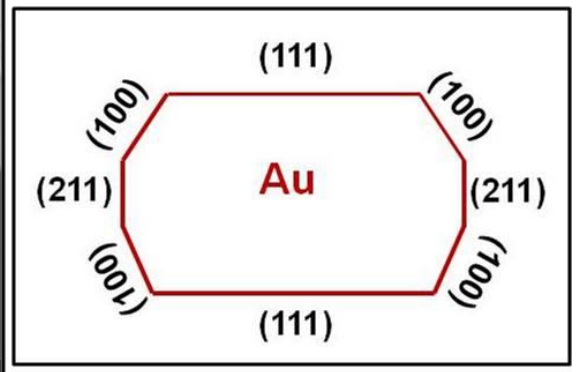
$= 0.3\times10^{-3} / 10^{-10}$

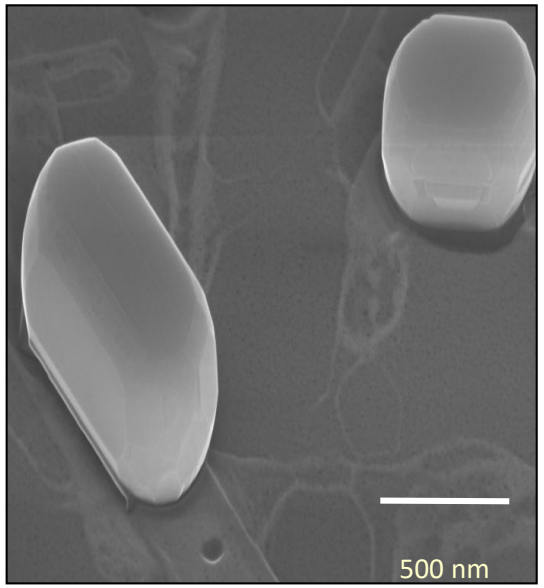
$= 3\times10^{6}$

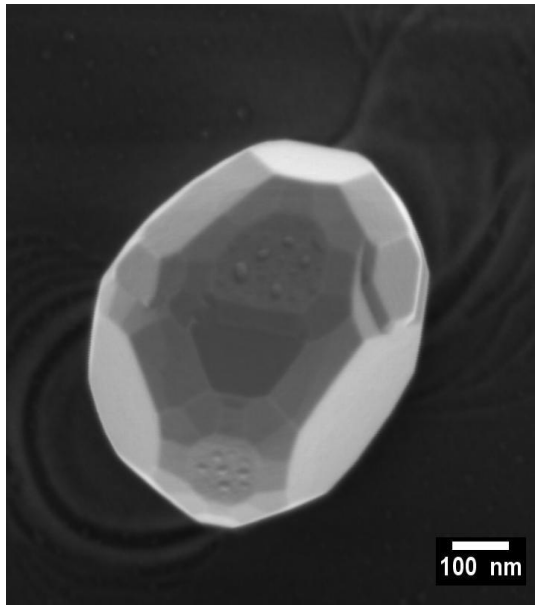
= 30,00,000 times

Growing Faceted Gold structures





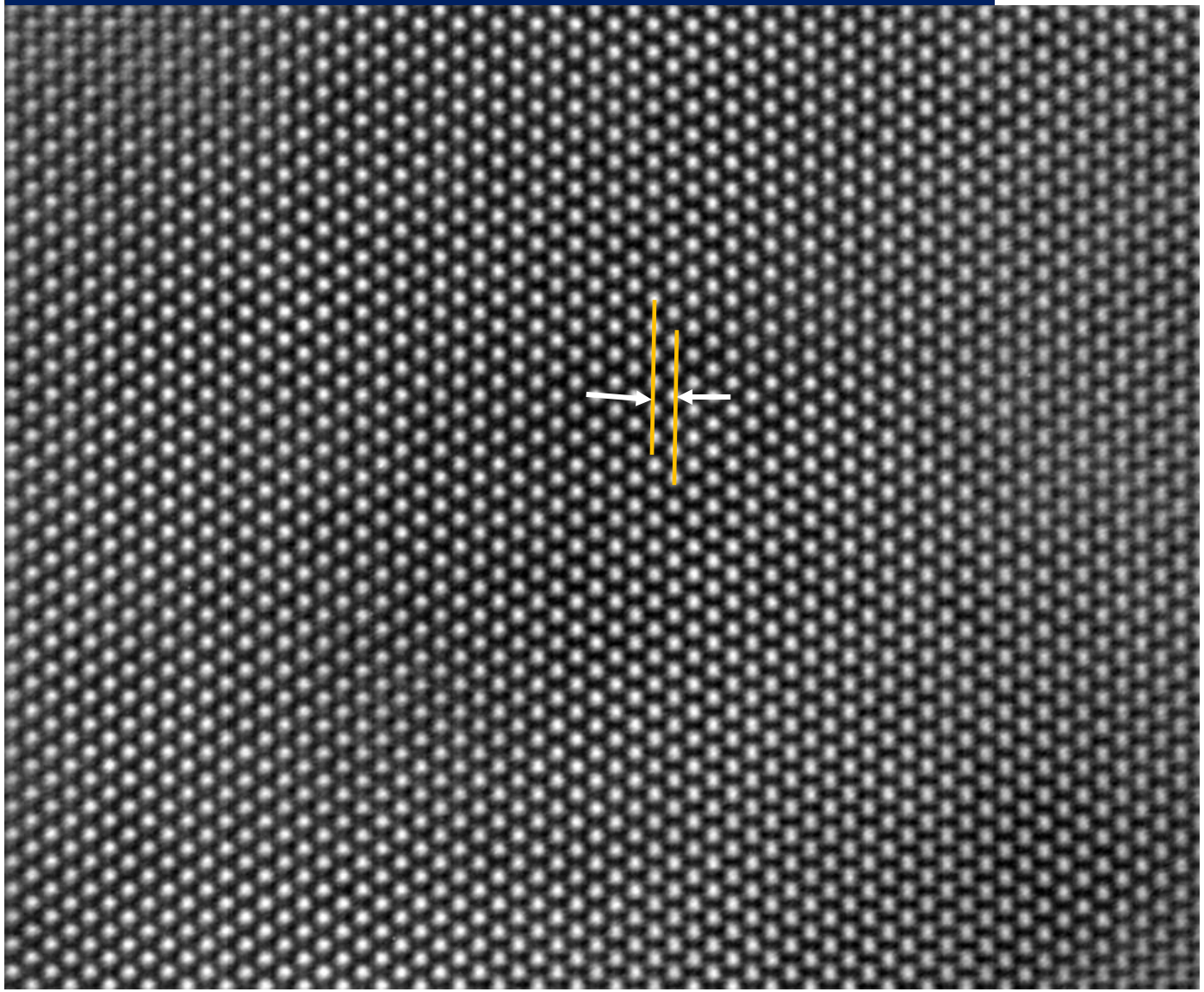




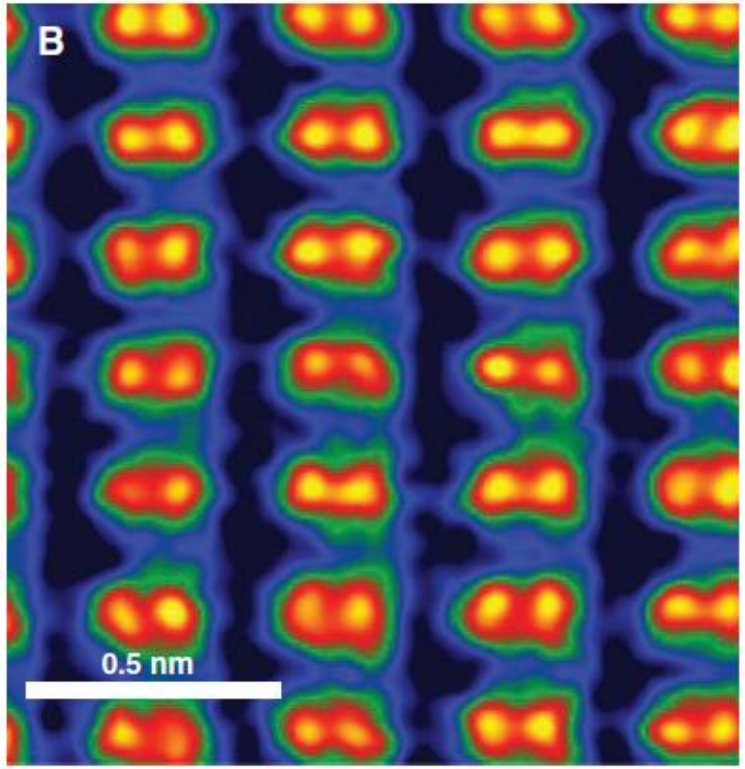
Rath, Raghav, Satyam
JAP (2013)

High Resolution TEM of Si single Crystal With 200 keV electrons @IOPB

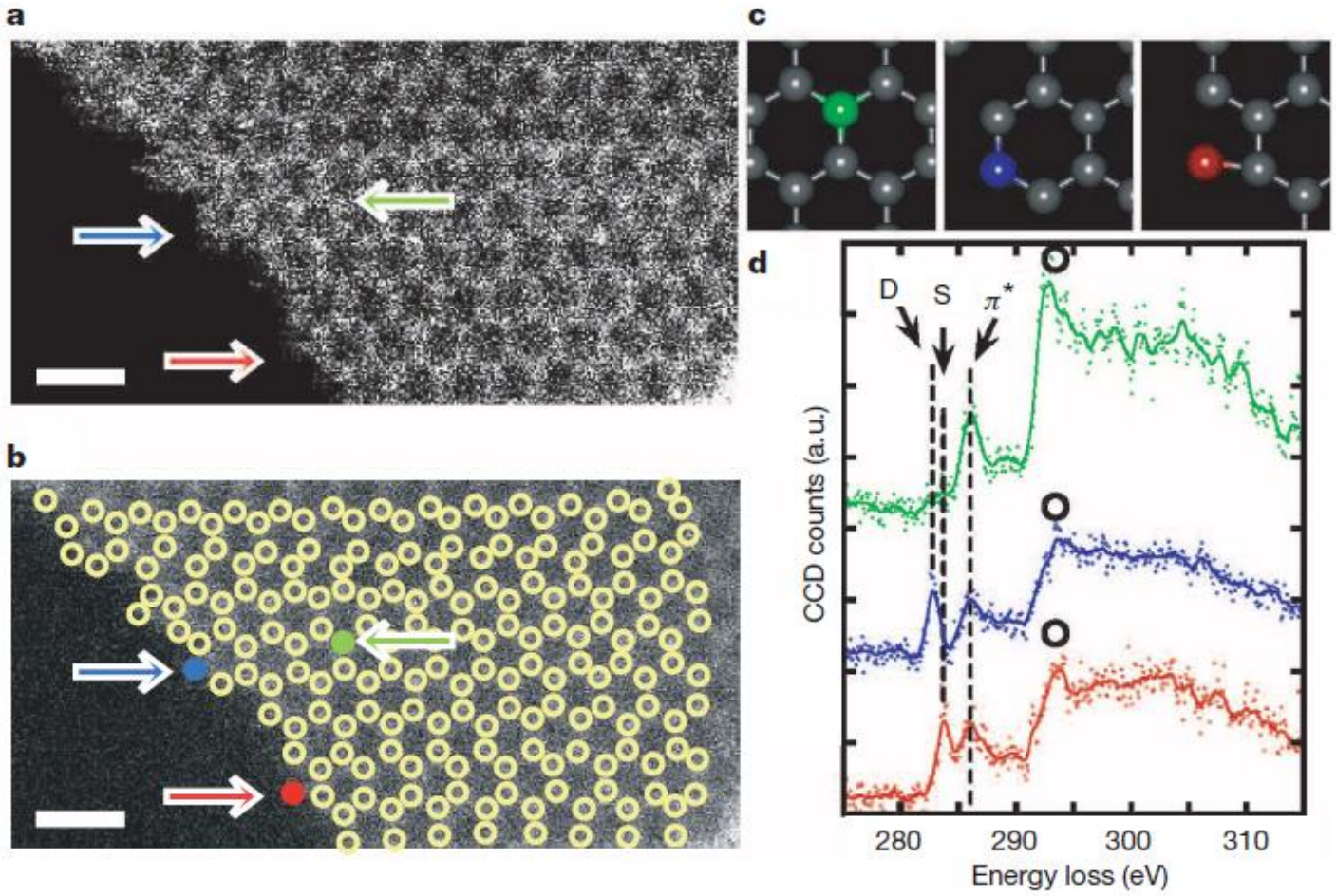
3.135 A



Aberration – Corrected TEM



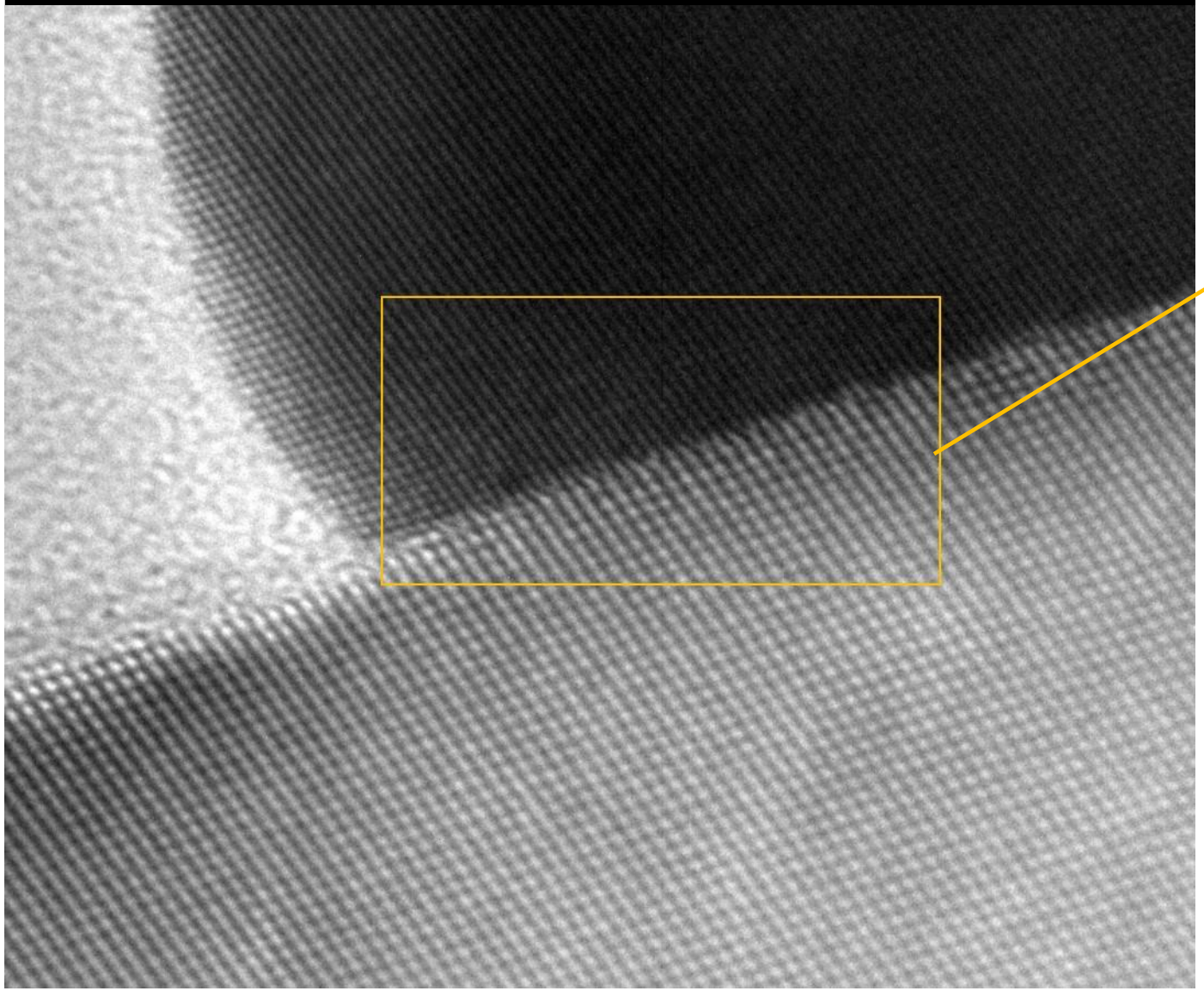
Si along a crystallographic [112] direction. The atom-pair separation of 78 pm is clearly resolved in the 300-keV instrument
K Urban, Science 321 (2008)

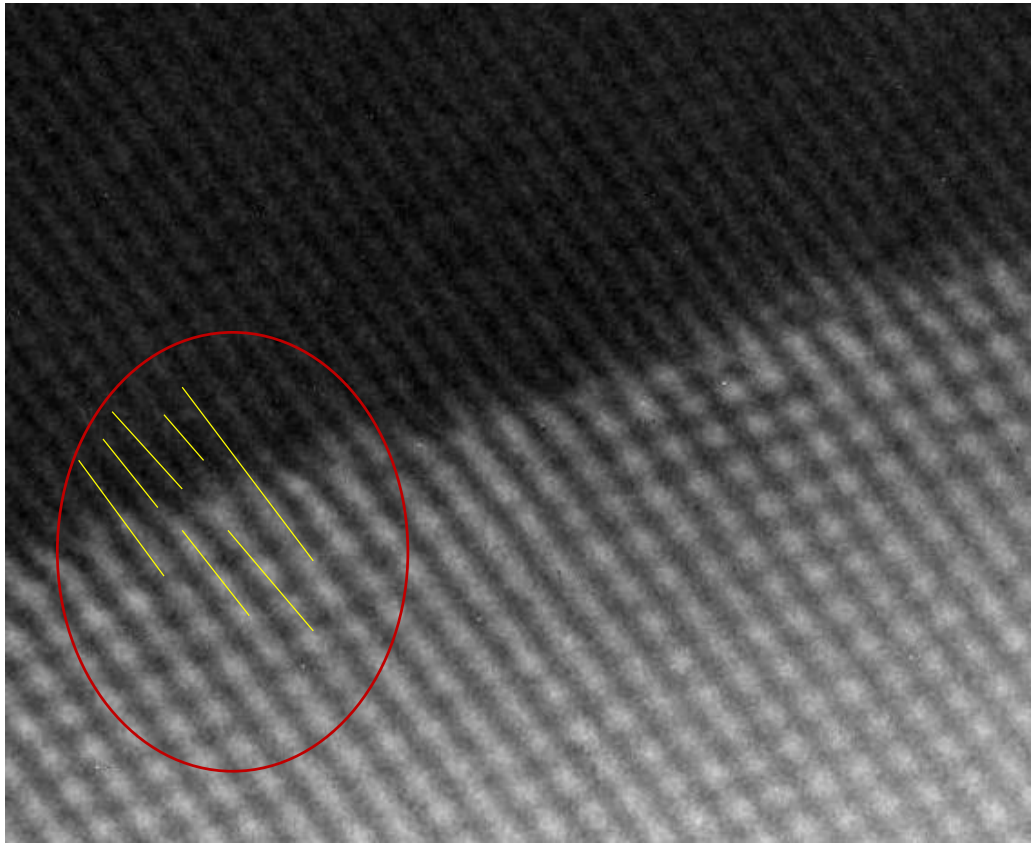


Atom –by-Atom Spectroscopy at Graphene Edge
Suenaga and Koshino , Nature 2010

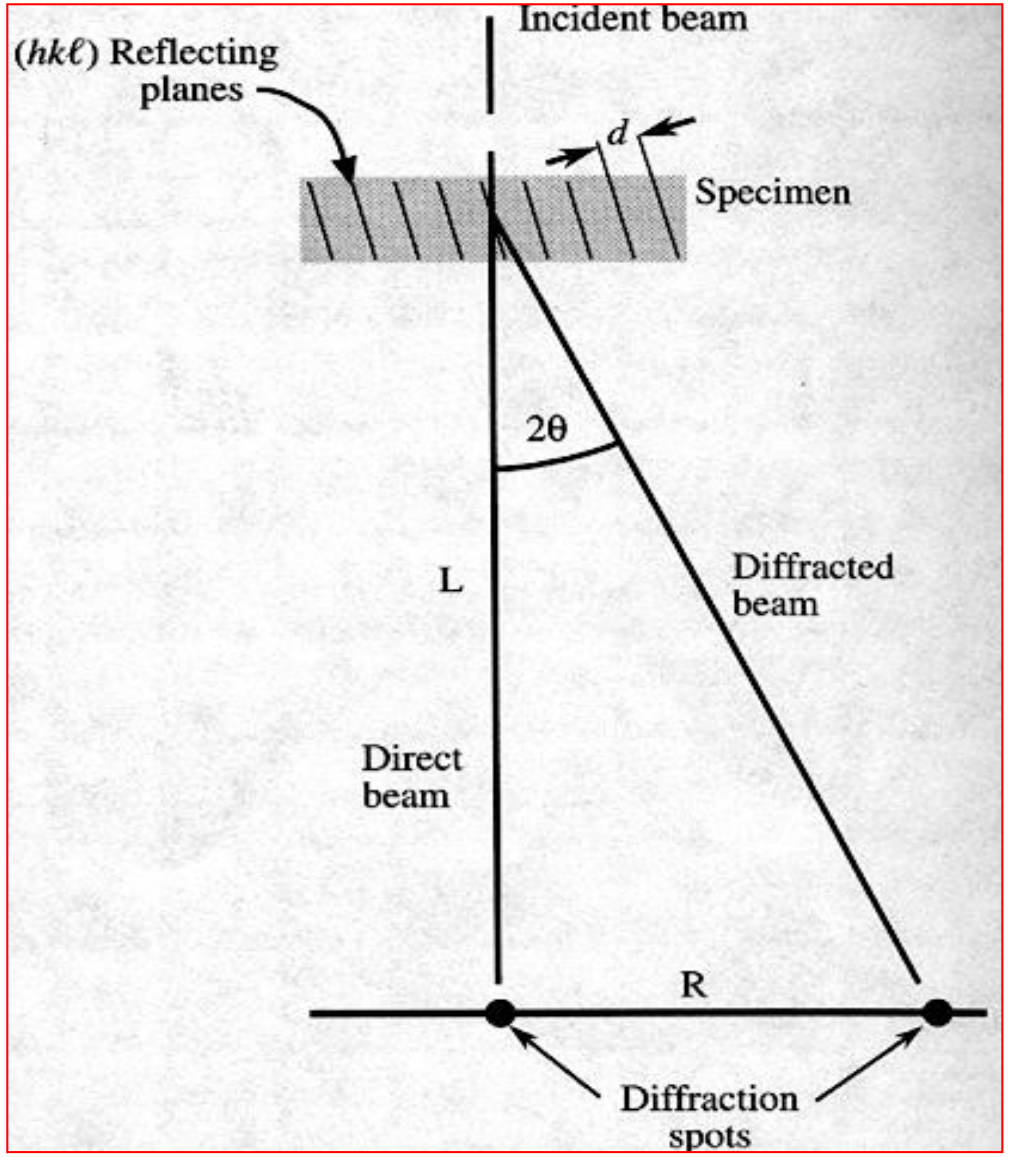
Can we look at Defects – like Dislocations?

Ag growth on Si(5 5 7) substrate





Determination of "d-spacing"



$2d \sin\theta = n\lambda$ (1)

$R/L = \tan 2\theta \approx 2\theta$ (2)

From above

$d = \lambda L/R$

For example:

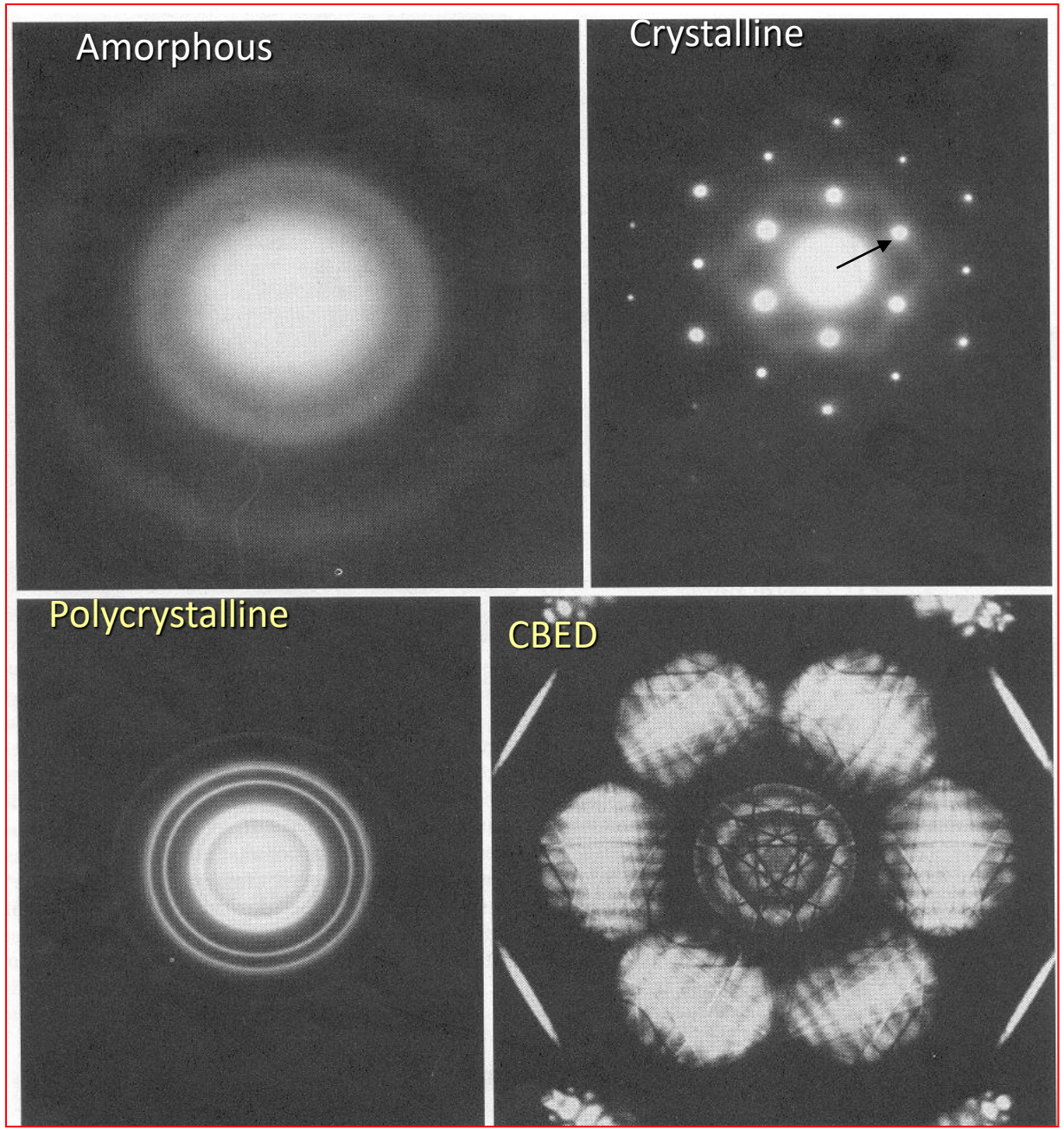
$d_{Si(111)} = 3.135$ Å

$\lambda = 0.025$ Å

L = 200 cm

Then: R = 1.59 cm

Diffraction : Examples



ELECTROMAGNETIC WAVES

Coupled electric and magnetic oscillations that move with the speed of light and exhibit typical wave behavior

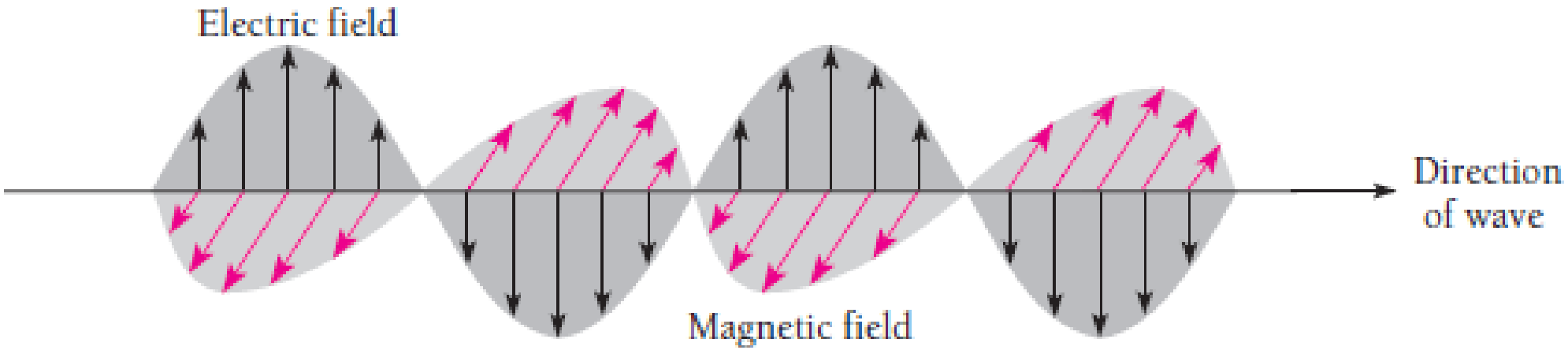


Figure 2.1 The electric and magnetic fields in an electromagnetic wave vary together. The fields are perpendicular to each other and to the direction of propagation of the wave.

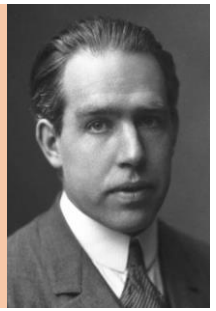
Is Quantum Mechanic Relevant for us NOW?

Is Quantum Mechanics a purely Abstract phenomena?

Your Future: What is Quantum Science and Technology?

- The Nobel Prize in Physics 1918 was awarded to Max Karl Ernst Ludwig Planck "in recognition of the services he rendered to the advancement of Physics by his discovery of energy quanta"

- The Nobel Prize in Physics 1922 was awarded to Niels Henrik David Bohr "for his services in the investigation of the structure of atoms and of the radiation emanating from them"



Discreteness of nature was further revealed in 1913 by Niels Bohr by introducing his model of atom. Bohr provided understanding of atom.

Hydrogen atom

Bohr model of atom

Planck Radiation:
All the discontinuities in nature are given in units based directly upon ***h***. The existence of this number and its particular size together form one of the great mysteries of nature. It appears explicitly or implicitly in every equation in quantum theory.

This is the revelation of new aspect of nature
The journey of Quantum Science has just begun.

Until the year 1900, we knew that LIGHT IS WAVE

But in 1900, Max Planck showed that LIGHT IS EMITTED IN PACKETS (QUANTA) OF ENERGY

1905, Albert Einstein established this DUAL NATURE OF LIGHT

When light Propagates it behaves like a WAVE
When light is EMITTED or ABSORBED, it behaves like PARTCLES

This is called Wave-Particle Duality

With all these knowledge since Max Planck's discovery of the quantum Nature of light, the rules (quantum mechanics) that small particles would obey, were formulated.

In the short period from 1925 to 1928, Heisenberg, Schrodinger, Born, Dirac, and many others laid the foundations of what is one of greatest theories of all the time, the theory of Quantum Mechanics. In generality and in range of application, it is unsurpassed. It has been so successful that one can not discuss atomic and nuclear matters without some understanding of this basic theory.